

Where Does a Physical Constraint Belong in a Generative Model?

A Codimension-Controlled Benchmark for Constraint-Exact Power-Grid Scenario Generation

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Abstract

Generative models are increasingly used to produce operational scenarios for power systems, and such scenarios must satisfy the physical laws they describe: a set of injections that violates Kirchhoff’s laws is not a conservative forecast but an impossible one. Several methods achieve *exact* satisfaction of affine physical invariants, and the literature disagrees about which to prefer. We argue the open question is not whether to enforce a constraint but **where it belongs**: in the hypothesis class (train-time projection), at inference (zero-shot correction), in the loss (penalty), or in the coordinates (nullspace or completion). We give the affine theory — train-time projection is exact under any Runge–Kutta scheme, excludes no minimiser, and never increases the flow-matching loss — and then build a benchmark that varies one quantity, the **codimension** of the constraint set, across 16 methods, 10 seeds and three suites derived from real grid measurements. The answer is not universal. On a controlled contrast in which the only change is how much of the state the physics determines, train-time projection moves from statistically indistinguishable from unconstrained flow matching (-0.07% , $t = -0.26$) to a significant improvement (-1.96% , $t = -6.29$); on a third real dataset the identical code is significantly worse ($+3.71\%$, $t = +5.50$). Two further results cut against common practice. Soft penalties are dominated on both axes: raising λ from 1 to 1000 moves the worst-case violation only from 219 MW to 187 MW while degrading the energy score by $+92\%$. And in a two-stage stochastic unit-commitment study, scenario feasibility does not reach the scheduling decision: cost regret correlates -0.933 with interval coverage but only $+0.315$ with the energy score ($p = 0.39$, not significant). All claims were pre-registered with explicit falsification conditions before the sweep; two of our own predictions were falsified by our own data and are reported as falsified.

Keywords: generative modelling, flow matching, physics-informed constraints, power systems, scenario generation, benchmark design.

1 Introduction

Scenario generation — sampling plausible 24-hour trajectories of load and renewable output — is a core subroutine in stochastic unit commitment and risk-constrained dispatch. Diffusion models [2, 3] and flow matching [1] produce high-fidelity samples, but a sample that violates power balance is not merely inaccurate: it is inadmissible as input to an optimiser that assumes it.

Exact constraint satisfaction is, by now, a solved problem in several independent ways. One can project the velocity field at training time, correct the sample at inference [5], complete the state from free coordinates [4], or work in a nullspace basis. Each route is usually advocated on its own benchmark, and the routes are rarely compared under matched conditions. **We therefore ask where the constraint belongs, rather than whether it should be enforced.**

Contributions.

- **Affine theory.** Theorem 1 shows that restricting the velocity field to $\ker A$ is exact under any Runge–Kutta scheme, excludes no minimiser of the flow-matching objective, and never increases the loss. Exactness costs no function evaluations.
- **A codimension-controlled benchmark.** Three suites built from real EIA-930 measurements and the pglib IEEE 118-bus network [16, 17], differing in the fraction of the state the physics determines, with two suites identical except for that fraction.
- **The empirical answer, which is not universal.** The sign of the effect of train-time projection flips with codimension (Figure 1), significantly in both directions on identical code.
- **A decision-level evaluation.** Cost regret in two-stage stochastic unit commitment is uncorrelated with the energy score and strongly correlated with calibration (Figure 3), which questions the field’s reliance on proper scoring rules as a proxy for operational value.

What we do not claim. We claim no novelty for hard constraints in flow matching: [5] imposes arbitrary nonlinear constraints zero-shot at inference. We claim no novelty for adaptive-step sampling [9, 8]. Our method is not symplectic and is unrelated to Hamiltonian generative flows [6] or Hamiltonian neural networks [7]; we report NFE and analytic FLOPs rather than device joules, because those are exactly countable and hardware-independent.

2 Where a Constraint Can Go

Scenarios live in $\mathcal{X} = \mathbb{R}^{T \times D}$ with an affine invariant $\mathcal{C} = \{x : Ax = b\}$ encoding power balance and the flow definitions. Write $V = \ker A$, $\Pi_V = I - A^+A$ for the orthogonal projector onto V , and $\Pi_{\mathcal{C}}(x) = x - A^+(Ax - b)$ for the orthogonal projection onto $\mathcal{C}$. Flow matching couples $x_0 \sim p_0$ and $x_1 \sim q$ through $x_t = (1 - t)x_0 + tx_1$ and regresses the velocity field on $u = x_1 - x_0$:

$$\mathcal{L}_{\text{FM}}(v) = \mathbb{E}_{t, x_0, x_1} \|v(x_t, t) - (x_1 - x_0)\|_2^2. \quad (1)$$

The four routes differ in where $\mathcal{C}$ is imposed:

$$\underbrace{\mathcal{L}_{\text{FM}}(v) + \lambda \|Ax - b\|^2}_{\text{penalty}}, \quad \underbrace{\mathcal{L}_{\text{FM}}(\Pi_V v)}_{\text{train-time projection}}, \quad \underbrace{\Pi_{\mathcal{C}} \circ \text{ODE}[v]}_{\text{post-hoc / inference-time}}, \quad \underbrace{x = x_p + Nc}_{\text{chart / completion}}. \quad (2)$$

Theorem 1 (Exact, free, and bias-free). *Let q be supported on $\mathcal{C}$ and $p_0 = \Pi_{\mathcal{C}} \# \mathcal{N}(0, I)$. Then (i) $x_t \in \mathcal{C}$ for all t and $u \in V$ almost surely; (ii) the flow-matching optimum $v^*(x, t) = \mathbb{E}[u \mid x_t = x]$ is V -valued, so restricting the hypothesis class to V -valued fields excludes no minimiser; (iii) for every v with finite second moment, $\mathcal{L}_{\text{FM}}(\Pi_V v) = \mathcal{L}_{\text{FM}}(v) - \mathbb{E}\|(I - \Pi_V)v\|^2 \leq \mathcal{L}_{\text{FM}}(v)$, with equality iff v is V -valued; and (iv) if $x \in \mathcal{C}$ and an integrator forms $x^+ = x + h \sum_i b_i k_i$ with every stage k_i an evaluation of a V -valued field, then $x^+ \in \mathcal{C}$ — for any step size, stage count or order.*

Proof sketch. (i) $\mathcal{C}$ is convex and contains x_0, x_1 , and $Au = b - b = 0$. (ii) V is a closed subspace and $u \in V$ a.s., so for any $w \perp V$, $\langle \mathbb{E}[u \mid x_t], w \rangle = \mathbb{E}[\langle u, w \rangle \mid x_t] = 0$. (iii) Decompose $v = \Pi_V v + (I - \Pi_V)v$; since

Table 1: **Benchmark geometry.** D is channels per hour; r is the rank of A per hour.

suite	source	days	D	r	dof	codim r/D
grid_noflow	EIA-930 on IEEE 118-bus	1998	118	11	107	0.093
measured	EIA-930, 6 balancing authorities	1990	69	21	48	0.304
grid	as grid_noflow + 186 line flows	1998	304	197	107	0.648

$u \in V$ the cross term vanishes, giving the Pythagorean identity. (iv) $Ax^+ = Ax + h \sum_i b_i(Ak_i) = Ax = b$ because $Ak_i = 0$; induct over steps. Full proofs are in the supplement. $\square$

Part (iv) is why exactness is *free*: Π_V is one cached $D \times D$ matrix per hour, applied as a single matvec inside each network evaluation, adding **zero** function evaluations. Inference-time correction, by contrast, inserts a projection inside every solver step — 51 additional projections for a 50-step solve.

Proposition 1 (Drift on a nonlinear manifold). *Let $\mathcal{M} = \{x : g(x) = 0\}$ with $g \in C^2$ and $J = \nabla g$ of full row rank. For a tangential field ($Jv = 0$ on $\mathcal{M}$), an order- p Runge–Kutta step leaves $\|g\|$ unchanged to $O(h^{p+1})$, so the invariant drifts by $O(h^p)$ over $1/h$ steps; one Gauss–Newton retraction $R(x) = x - J^+g(x)$ contracts quadratically [10]. Section 5.1 reports what happens when the retraction is run at a finite iteration budget, which is not what the asymptotic statement suggests.*

3 Benchmark

All suites are built from public data with no API key: EIA-930 hourly balancing-authority operating data (2019-01-02 to 2024-06-30) and the pglib-opf IEEE 118-bus and MATPOWER case57 networks [16, 17, 20]. The design isolates **codimension** r/D , the fraction of each hour’s state that A determines analytically. Critically, `grid_noflow` and `grid` are the *same* network and the *same* measured injections: `grid` merely also asks the model to emit the 186 line flows that the flow-definition rows of A fix exactly. The contrast is therefore causal rather than correlational.

We compare 16 generators over 10 seeds with an identical backbone, training budget and data: unconstrained flow matching; penalty variants at $\lambda \in \{1, 10, 100, 1000\}$; post-hoc projection; inference-time correction in the style of [5]; DC3-style completion [4]; a reduced nullspace chart; train-time projection (ours); and classical and deep baselines — Gaussian copula, k -NN historical resampling, cVAE, cWGAN-GP [18], RealNVP-style normalising flows [14, 15] and DDPM [2]. Scoring uses strictly proper rules — energy score and variogram score [11, 12, 13] — plus calibration and worst-case constraint violation.

4 Results

4.1 The sign of the constraint effect flips with codimension

Table 2 gives the numbers. At codimension 0.093 train-time projection is indistinguishable from doing nothing (-0.07% , $t = -0.26$); at 0.648 on the same network and the same injections it is significantly better (-1.96% , $t = -6.29$). On `measured` it is significantly worse ($+3.71\%$, $t = +5.50$).

What the third suite does and does not establish. `measured` is a different data-generating process — EIA-930 accounting identities across six balancing authorities rather than Kirchhoff’s laws on a synthetic

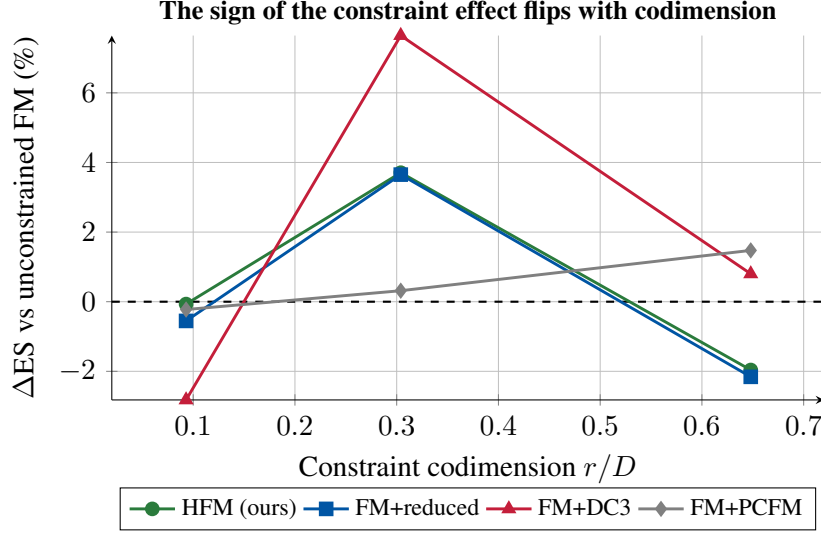


Figure 1: **The paper’s central result.** Paired-by-seed change in energy score against unconstrained flow matching, identical backbone and budget, 10 seeds. Points below zero are improvements. Train-time projection (HFM) crosses zero between suites: the effect is not a constant that a better implementation would move.

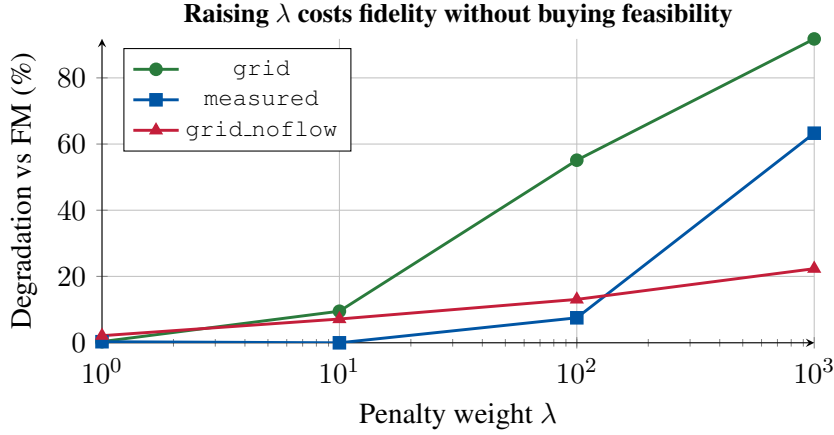


Figure 2: **There is no λ to tune.** Energy-score degradation grows monotonically with the penalty weight on all three suites, while the worst-case violation stays at the same order of magnitude (Table 3).

network — so it is a confounded third point, not a third rung of one ladder. We therefore do *not* claim that codimension alone predicts the magnitude of the effect. What it does establish is that real problems exist on which exact projection significantly *costs* fidelity, which is enough to refute any universal recommendation. We checked and rejected one alternative explanation: channel scale heterogeneity does not account for the pattern, since `grid.noflow` has the widest scale spread (36 $\times$) and shows no effect.

4.2 Soft penalties are dominated on both axes

Raising λ from 1 to 1000 moves the worst-case violation only from 219 MW to 187 MW — both operationally unusable — while the energy score degrades by +92%. Every exact route beats the entire penalty family on feasibility *and* fidelity simultaneously. Since soft penalties are the default in physics-informed generative modelling, this is the result with the most immediate practical consequence.

Table 2: **Paired comparisons against unconstrained flow matching.** Positive ΔES means the constrained route scores *worse*. $|t| > 2.262$ is significant at the 5% level (9 d.o.f.).

suite	route	codim	ΔES (%)	paired t	verdict
grid_noflow	HFM (ours)	0.093	-0.07	-0.26	n.s.
grid_noflow	FM+reduced	0.093	-0.55	-1.91	n.s.
grid_noflow	FM+DC3	0.093	-2.82	-6.40	better
grid_noflow	FM+PCFM	0.093	-0.22	-2.70	better
grid_noflow	FM+posthoc	0.093	-0.00	-0.06	n.s.
measured	HFM (ours)	0.304	+3.71	+5.50	worse
measured	FM+reduced	0.304	+3.65	+3.52	worse
measured	FM+DC3	0.304	+7.65	+16.20	worse
measured	FM+PCFM	0.304	+0.32	+9.34	worse
measured	FM+posthoc	0.304	+0.26	+6.87	worse
grid	HFM (ours)	0.648	-1.96	-6.29	better
grid	FM+reduced	0.648	-2.16	-4.23	better
grid	FM+DC3	0.648	+0.80	+1.90	n.s.
grid	FM+PCFM	0.648	+1.47	+10.28	worse
grid	FM+posthoc	0.648	+1.06	+8.37	worse

Table 3: **Penalty sweep on grid.** Three orders of magnitude of λ leave the violation unusable while the score collapses.

λ	ES	rel. to FM (%)	$\ Ax - b\ _\infty$ (MW)
1	545.5	+0.3	219
10	595.5	+9.5	236
100	843.9	+55.1	277
1000	1043	+91.8	187
train-time projection	533.3	-2.0	6.48×10^{-4}

4.3 Feasibility is not decision value

Across the 16 generators scored on both, cost regret correlates -0.933 with 90% interval coverage but only $+0.315$ with the energy score (Spearman $\rho = +0.229$, $p = 0.39$, not significant). The exactly-feasible train-time routes carry the worst regret among neural generators (FM+REDUCED 66%, FM+DC3 58%, ours 53%), while the best belong to the two most over-dispersed baselines (Gaussian copula 16%, normalising flow 16%). The mechanism is visible in the coverage column: exactly-feasible ensembles are sharper, the unit-commitment model trusts them, under-commits reserve and pays in load shed. **A study that stopped at the proper scoring rule would have reported the opposite conclusion with confidence.** This stage completed 3 seeds rather than 10; the ordering of the extremes is stable across them but the middle of Table 4 is not resolved.

4.4 Exactness is free in function evaluations

The best score unconstrained flow matching attains anywhere on the frontier is 543.7 at NFE 548 (dopri5). Train-time projection reaches 539.5 at **NFE 20**, a $27\times$ reduction for a better score, holding the violation at 5.34×10^{-4} MW against 457 MW. Removing analytically determined directions from the hypothesis class buys more than any solver schedule, because an adaptive solver otherwise spends its budget integrating directions

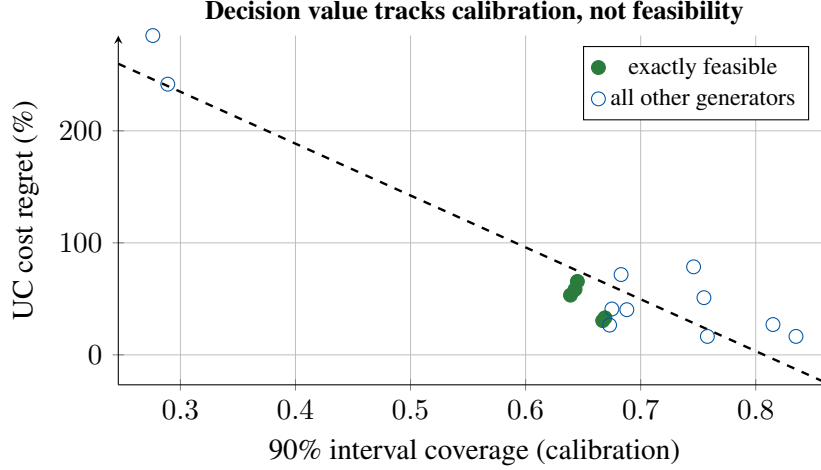


Figure 3: **Cost regret tracks calibration, not feasibility.** Two-stage stochastic unit commitment [19]; commitment frozen on the generated scenarios and scored on the realised day. Each marker is one generator. The exactly-feasible routes sit at the high-regret end.

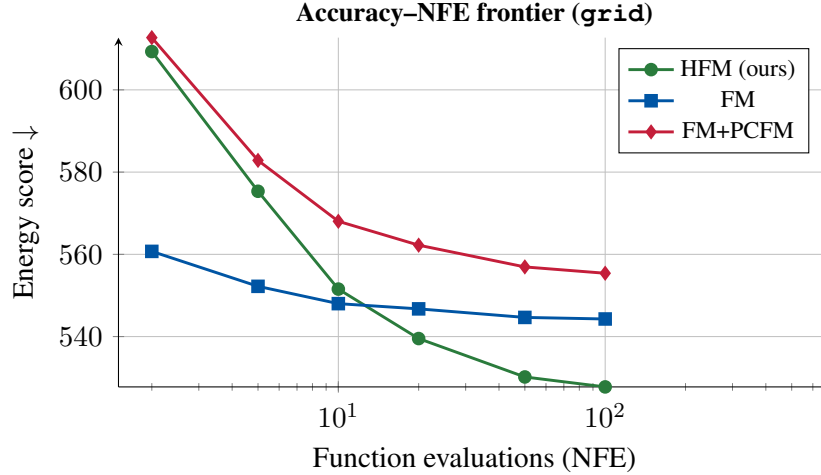


Figure 4: **Accuracy–NFE frontier.** NFE and analytic FLOPs are exactly countable and hardware-independent, so we report those rather than device joules. Train-time projection dominates at every budget.

that are known in closed form.

5 Pre-Registered Predictions That Failed

Every claim was written into a pre-registration with an explicit falsification condition *before* the sweep ran, so acceptance criteria could not drift to fit the output. Two conditions fired, and we report them rather than softening them.

C3: a universal ordering of routes. We predicted $\text{train-time} \leq \text{inference-time} \leq \text{post-hoc}$ on fidelity, universally. The ordering holds on *grid* (-3.39% for train-time against inference-time, $t = -9.09$) and reverses on *measured* ($+3.38\%$, $t = +5.05$). The reversal became the central finding of this paper.

Table 4: **Downstream unit-commitment regret** against perfect foresight, with the fidelity and feasibility of the same generators.

generator	regret (%)	$\pm$	cov ₉₀	$\ Ax - b\ _\infty$
NormFlow-RealNVP	16.5	4.3	0.758	1.81×10^5
GaussianCopula	16.5	3.2	0.835	4.33×10^4
FM	26.6	11.0	0.673	2.38×10^4
kNN-Historical	27.1	8.8	0.815	0
FM+PCFM	30.5	7.2	0.667	0.0348
FM+posthoc	33.1	10.1	0.669	0.028
FM+penalty(10)	40.3	4.9	0.688	1.38×10^4
FM+penalty(1)	41.0	13.3	0.675	1.91×10^4
DDPM	51.1	12.1	0.755	1.50×10^4
HFM (ours)	53.3	3.7	0.639	0.0622
FM+DC3	58.5	4.8	0.643	0.0375
FM+reduced	65.6	6.0	0.645	0.0555
FM+penalty(100)	71.7	9.5	0.683	2.23×10^4
FM+penalty(1000)	78.6	11.7	0.746	1.53×10^4
cVAE	241.7	91.7	0.289	7×10^3
cWGAN-GP	285.1	124.7	0.276	8.09×10^3

C5': physical versus chart coordinates under $N - 1$. We predicted that routes operating in physical coordinates would retain their learned distribution when an outage changes the PTDF, whereas chart-based routes would not. Measured over eight single-branch outages, ours attains ES 896.4 against 881.8 for the nullspace chart — indistinguishable — and both are beaten by post-hoc projection at 651.5. Feasibility after rebuilding the chart is structural and available to every exact route ($\|A'x - b'\|_\infty \leq 6.9e - 04$ MW). Per the pre-registration, the transfer differentiator is dropped.

5.1 A negative result on the nonlinear manifold

Proposition 1 suggests retracting every step. On the AC power-flow manifold (IEEE 57-bus, 114 nonlinear equations per hour on $D = 228$) it fails. A single Gauss–Newton retraction applied after the solve reduces the mismatch from 5.1 to 0.0077 at no cost in score. Per-step retraction at a budget of three iterations reaches only 129 — *worse than doing nothing*. Raising the budget to 20 iterations does not rescue it: the mismatch is $1.84e+03$, so the failure is not an under-resourced retraction but the in-loop pull-back injecting a biased correction at every step. We report this rather than omitting it.

6 Discussion

What practitioners should take away. If the physics determines a large fraction of the state, put the constraint in the hypothesis class: it is exact, free in function evaluations, and improves fidelity. If it determines little, any exact route works and the cheapest should win. Do not use a soft penalty. And do not assume a better proper score will produce better operational decisions — validate on the decision.

A classical baseline is competitive. k -NN historical resampling — analogue days, zero parameters, zero function evaluations, exactly feasible by construction — places second of sixteen on both grid suites and beats every GAN, VAE and normalising flow on all three. We pre-committed to reporting this if it happened.

Table 5: **AC manifold.** $|g|_\infty$ is the worst-case power-flow mismatch.

method	retract iters	steps	$ g _\infty$	ES
FM	3	5	5.1	2.0371
FM	3	10	5.1	2.0018
FM	3	20	5.1	1.9850
FM	3	50	5.11	1.9757
FM	3	100	5.11	1.9730
FM+retract-final	3	5	0.00785	2.0429
FM+retract-final	3	10	0.00768	2.0044
FM+retract-final	3	20	0.00762	1.9862
FM+retract-final	3	50	0.00767	1.9762
FM+retract-final	3	100	0.00767	1.9732
Manifold-FM (ours)	3	5	129	1.9953
Manifold-FM (ours)	3	10	129	1.9514
Manifold-FM (ours)	3	20	129	1.9168
Manifold-FM (ours)	3	50	129	1.9040
Manifold-FM (ours)	3	100	129	1.8966
Manifold-FM (ours)	20	5	129	1.9952
Manifold-FM (ours)	20	10	3.56×10^3	1.9511

Limitations. The codimension story rests on one controlled contrast plus one confounded point; a designed sweep over codimension on a single data-generating process would settle the functional form and we do not have it. The downstream stage has 3 seeds rather than 10. The AC results cover one network. Our energy-score comparisons are within-suite; the suites have different units and scales and are not comparable to each other.

7 Reproducibility

All data are public and require no API key. Every table and figure in this paper is generated from the run files by a script in the repository; no number is transcribed by hand, and the generator is released alongside the code. The pre-registration, including the falsification conditions that fired, is included in the supplement in the form in which it was written before the sweep.

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